How do i detect the MicroSD card adapter, which works thru SDMMC, to make sure that the code can check presence or non presense?

Detecting MicroSD Card Presence via SDMMC

There are two main mechanisms to detect SD card presence when using the SDMMC host controller on ESP32.

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1. Hardware Card Detect (CD) Pin

The SDMMC controller has a dedicated `card_detect_n` input signal. You configure this in the `sdmmc_slot_config_t` structure:

```
sdmmc_slot_config_t slot_config = SDMMC_SLOT_CONFIG_DEFAULT();
slot_config.cd = GPIO_NUM_X; // Replace with your actual CD GPIO pin
slot_config.wp = SDMMC_SLOT_NO_WP; slot_config.width = 4;
```

If no CD pin is wired, the driver defaults to "card present" (pulls the signal low internally):

```
slot_config.cd = SDMMC_SLOT_NO_CD; // Assumes card always present
```

The hardware register `cdetect` reflects the raw state — **bit N reads 0 if card N is present** (active-low logic).

SD/MMC Signals

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2. Software Status Check via `sdmmc_get_status()`

After mounting, you can poll card presence programmatically using `sdmmc_get_status()`. This is what the FATFS diskio layer uses internally:

```
esp_err_t err = sdmmc_get_status(card); if (err != ESP_OK) { // Card is not
present or has an error ESP_LOGE(TAG, "Card not available (0x%x)", err); }
else { // Card is present and responding }
```

diskio_sdmmc.c

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3. GPIO-Based Detection (Manual)

If your adapter has a physical card-detect switch wired to a GPIO (common on many breakout boards), you can read it directly:

```
const gpio_config_t sd_detect_config = { .pin_bit_mask =
BIT64(YOUR_CD_GPIO), .mode = GPIO_MODE_INPUT, .pull_up_en =
GPIO_PULLUP_DISABLE, .pull_down_en =
GPIO_PULLDOWN_DISABLE, .intr_type = GPIO_INTR_DISABLE };
gpio_config(&sd_detect_config); bool card_present =
(gpio_get_level(YOUR_CD_GPIO) == 0); // Active-low
```



esp32_p4_eye.c

Summary

Method	When to Use
`slot_config.cd = GPIO_NUM_X`	Adapter has a physical CD pin wired to a GPIO
`sdmmc_get_status()`	Poll presence after init (software check)
Direct `gpio_get_level()`	Custom GPIO-based card detect circuit

The most reliable approach is to **combine** a hardware CD pin (if available on your adapter) with periodic `sdmmc_get_status()` polling to detect both insertion and removal events.

So, I still need to install the sdmmc initialization functions first, and then I can use the get_status() calls?

Yes, that's correct. You must complete the full initialization sequence before `sdmmc_get_status()` can be used. Based on the knowledge sources, the required order is:

1. **`sdmmc_host_init()`** — initializes the SDMMC host peripheral
2. **`sdmmc_host_init_slot()`** — configures the slot and GPIO pins
3. **`sdmmc_card_init()`** — probes and initializes the card, populates the `sdmmc_card_t` structure

Only after all three steps succeed can you call `sdmmc_get_status()`, since it requires a valid, initialized `sdmmc_card_t` pointer.



Protocol Layer API

A typical pattern looks like this:

```
sdmmc_host_t host = SDMMC_HOST_DEFAULT(); sdmmc_slot_config_t
slot_config = SDMMC_SLOT_CONFIG_DEFAULT(); slot_config.cd =
YOUR_CD_GPIO; // optional hardware CD pin sdmmc_card_t *card =
malloc(sizeof(sdmmc_card_t)); // Step 1: Init host
ESP_ERROR_CHECK(sdmmc_host_init()); // Step 2: Init slot
ESP_ERROR_CHECK(sdmmc_host_init_slot(SDMMC_HOST_SLOT_1,
&slot_config)); // Step 3: Init card esp_err_t ret = sdmmc_card_init(&host,
card); if (ret != ESP_OK) { ESP_LOGE(TAG, "Card not found or init failed:
%s", esp_err_to_name(ret)); } // Now safe to poll status esp_err_t status =
sdmmc_get_status(card); if (status != ESP_OK) { ESP_LOGE(TAG, "Card
removed or error"); }
```

tusb_msc_main.c

One important note: **`sdmmc_card_init()` itself will fail if no card is present**, so it also serves as an initial presence check. After that, `sdmmc_get_status()` is useful for detecting card removal during runtime.

SDMMC Functions